Characterizing the MSR 2026 AI Development Dataset: Structure, Quality, and Metadata Analysis

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***Abstract*—The MSR 2026 Challenge dataset contains 42,179 pull requests labeled with AI agent attribution from the official MSR 2026 HuggingFace PR-level release (aidata.csv). This paper characterizes this dataset through systematic analysis of all records in the dataset. We document real agent distribution: GitHub Copilot 87.8% (37,042 PRs) and Claude Code 12.2% (5,137 PRs). Analysis reveals 1,934 unique users with average 21.8 PRs per user, high completeness of core text fields (99.3%, though label validity and PR authenticity remain unverified), and 81.7% closure rate. The top contributor accounts for 36,596 PRs while median user contributes 1 PR. This characterization enables informed empirical research design and methodological decisions for the MSR 2026 Challenge dataset.**

***Index Terms*—dataset analysis, descriptive statistics, data quality, pull requests, MSR challenge, empirical software engineering**

I. Introduction

Large-scale software engineering datasets require rigorous characterization before use in empirical research. The official MSR 2026 HuggingFace PR-level release (aidata.csv), containing **42,179 pull requests** with AI agent labels, presents valuable opportunities for dataset-level descriptive analysis and characterization.

This paper provides a comprehensive characterization of the dataset structure based on genuine analysis of the official MSR 2026 HuggingFace PR-level release (aidata.csv). Through systematic analysis of all 42,179 records, we document the distributional properties, user patterns, and data quality characteristics that inform empirical research design.

Our analysis reveals important structural properties: PRs labeled as **'GitHub Copilot' account for 87.8% of records** (37,042 PRs), while **'Claude Code' labels represent 12.2%** (5,137 PRs). However, because 86.7% of all PRs originate from a single system-associated account, these records cannot be assumed to represent independent tool usage. The dataset spans **1,934 unique users** with substantial contribution variation—from single-PR contributors (median) to one user contributing 36,596 PRs.

A notable characteristic of this dataset is the extreme contribution concentration: one user account contributes 36,596 of the 42,179 pull requests (86.7%). This concentration likely reflects automation or institutional data collection rather than individual developer behavior, and must be considered when designing empirical studies. Understanding these distributional patterns is essential for appropriate statistical analysis and methodological design.

This work analyzes the MSR 2026 Challenge dataset. Research conducted at Edinburgh Napier University, November 2025.

A. Motivation

Large-scale datasets require careful characterization before analysis. Understanding dataset properties is crucial for:

- **Methodological Design**: Researchers need to understand data limitations
- **Sampling Strategies**: Distribution imbalances affect statistical analysis
- **Quality Assessment**: Data anomalies must be documented
- **Baseline Establishment**: Descriptive statistics support future work

B. Research Questions

We systematically characterize the MSR 2026 dataset through four research questions:

RQ1: *What are the structural and distributional properties of the dataset, including label imbalances and user patterns?*

RQ2: *What data quality issues exist, including missing fields and metadata completeness patterns?*

RQ3: *What are the distributions of key metadata fields (title lengths, description availability, closure rates) and how do they vary across labels?*

RQ4: *What methodological constraints and preprocessing requirements emerge from the dataset's structural and quality characteristics?*

C. Key Contributions

This paper makes several descriptive contributions:

1) **Empirical dataset characterization** documenting actual distributions and metadata patterns across all 42,179 pull requests from aidata.csv
2) **Observed statistical analysis** computing agent distributions, user patterns, and data quality metrics
3) **Directly computed baselines** providing verified reference statistics for future research
4) **Methodological documentation** based on actual dataset properties
5) **Replication materials** with code and visualizations derived from real data

II. Related Work

A. Dataset Characterization in Software Engineering

Software engineering research increasingly relies on large-scale datasets that require systematic characterization before use. Bird et al. [5] demonstrated how dataset biases affect conclusions about code ownership and quality. Kim et al. [6] showed that software change datasets contain systematic labeling issues that impact classification validity.

Recent work has emphasized the importance of understanding dataset limitations before drawing empirical conclusions. These studies establish the methodological foundation for our dataset characterization approach.

B. Data Quality in Mining Software Repositories

MSR research faces persistent challenges with data quality and validity. Bacchelli and Bird [7] documented how GitHub data contains systematic biases that affect empirical conclusions. Their work highlights the need for comprehensive dataset characterization before behavioral analysis.

Studies of large-scale software repositories consistently find missing data, labeling inconsistencies, and structural biases that threaten research validity [8]. Our work follows this tradition by systematically documenting quality issues before they impact research conclusions.

C. Methodological Issues in Observational Studies

Inozemtseva and Holmes [9] demonstrated how observational software engineering data can lead to spurious correlations when structural biases are not addressed. Their analysis shows the importance of understanding data generation processes before making causal claims.

This methodological perspective motivates our focus on dataset characterization rather than behavioral inference. Understanding data limitations enables more rigorous empirical research design.

III. Methodology

A. Dataset and Analysis Scope

We analyzed the official MSR 2026 HuggingFace PR-level release (aidata.csv) containing **42,179 pull requests** (147.3 MB). The dataset includes fields for agent labels, user IDs, timestamps, PR titles, descriptions, closure status, and repository metadata across 14 columns.

Our analysis processed the entire dataset to capture structural patterns and distributional properties. We used pandas for data manipulation and matplotlib/seaborn for visualization generation, producing real visualizations documenting actual dataset characteristics computed from the aidata.csv file.

B. Real Dataset Distributions

The dataset shows a two-agent distribution:

- **GitHub Copilot**: 37,042 PRs (87.8%)
- **Claude Code**: 5,137 PRs (12.2%)

This binary distribution reflects the specific AI agents captured in the MSR 2026 dataset, with GitHub Copilot representing the dominant presence. While both labels are numerically represented, extreme user-level clustering severely limits any valid comparative behavioral analysis between agents.

TABLE I
MSR 2026 Dataset Overview

Characteristic	Value
Total Pull Requests	42,179
GitHub Copilot	37,042 (87.8%)
Claude Code	5,137 (12.2%)
Unique Users	1,934
Top User Contribution	36,596 PRs (86.7%)
Median User Contribution	1 PR
Data Completeness	99.3%
Closure Rate	81.7%

C. Data Quality Assessment Methods

We systematically assessed data quality through multiple approaches:

User Pattern Analysis: Examined user ID distributions per agent to identify suspicious patterns like single-user dominance or bot accounts.

Metadata Distribution Analysis: Computed descriptive statistics for title lengths, body lengths, and closure rates across agent categories.

Structural Assessment: Analyzed label imbalances, category sizes, and user-to-PR ratios to identify potential data collection artifacts.

D. Metadata Characterization Methods

We computed comprehensive descriptive statistics for all metadata fields:

Text Field Analysis: Analyzed title and description lengths, computed character count distributions, identified encoding issues and unusual content patterns.

Statistical Analysis: Computed comprehensive descriptive statistics including user distributions, agent ratios, closure rates, and missing data patterns.

Visualization Generation: Created visualizations documenting agent distributions and user activity patterns using matplotlib and seaborn for data characterization.

IV. Results

A. RQ1: Structural and Distributional Properties

Analysis of the dataset reveals clear distributional properties based on real data:

Binary Agent Distribution: The dataset contains two AI agents with a moderate class imbalance (7.2:1): GitHub Copilot with 37,042 PRs (87.8%) and Claude Code with 5,137 PRs (12.2%). While both labels are numerically represented, extreme user-level clustering severely limits any valid comparative behavioral analysis between agents.

Observed Agent Label Distribution visualization available in the replication repository: https://github.com/ghost1100/MSR

User Distribution Analysis: The complete dataset spans **1,934 unique users** with extreme contribution concentration:

TABLE II
AGENT DISTRIBUTION SUMMARY

Agent	Total PRs	Percentage	Closed	Open
GitHub Copilot	37,042	87.8%	30,313	6,729
Claude Code	5,137	12.2%	4,152	985
Total	**42,179**	**100.0%**	**34,465**	**7,714**

TABLE III
USER CONTRIBUTION STATISTICS

Metric	Value
Total unique users	1,934
Mean PRs per user	21.8
Median PRs per user	1.0
Max PRs (single user)	36,596
Top user percentage	86.7%
Users with 1 PR only	1,689 (87.3%)

- **Average contributions**: 21.8 PRs per user with extreme variance
- **Median contributions**: 1.0 PR indicating many single-contribution users
- **Extreme concentration**: One user ("Copilot") contributed 36,596 PRs (86.7% of dataset)
- **Distribution pattern**: Highly skewed indicating potential automated generation or bot activity

User Concentration Analysis: The extreme dominance of a single user requires careful interpretation. The "Copilot" user account may represent:

- Automated or institutionally generated PR activity associated with the Copilot label
- A bot or system account rather than human developer activity
- Batch data collection artifacts requiring methodological consideration

B. RQ2: Data Quality Assessment

Empirical data quality analysis of the dataset reveals good field-level completeness with specific considerations. We did not observe systematic timestamp anomalies or large-scale duplicate records in the dataset:

Missing Data Assessment: The dataset demonstrates high completeness across key fields:

- **Missing titles**: 1 record (0.002% of 42,179 total)
- **Missing body content**: 300 records (0.71% of dataset)
- **Empty fields**: No empty titles or body content detected
- **Overall field completeness**: 99.3% complete data coverage (though label validity remains unverified)

User Contribution Distribution: Analysis reveals characteristic patterns in real-world usage:

- **Contribution concentration**: One user contributed 36,596 PRs (86.7% of dataset)
- **Median user activity**: 1 PR per user indicates many occasional contributors
- **User diversity**: 1,934 unique users across both AI agents

TABLE IV
DATA COMPLETENESS SUMMARY

Field	Missing	Complete
Title	1 (0.002%)	42,178 (99.998%)
Body	300 (0.71%)	41,879 (99.29%)
User	0 (0%)	42,179 (100%)
Agent	0 (0%)	42,179 (100%)
State	0 (0%)	42,179 (100%)
Overall	**301 (0.36%)**	**99.64%**

- **Distribution characteristics**: Highly skewed toward few high-volume contributors

While skewed contribution distributions are common in open-source projects, the extreme dominance observed here (86.7% from one account) exceeds typical human contribution patterns and likely reflects automation or institutional data collection rather than individual developer behavior.

C. RQ3: Metadata Characteristics and Patterns

Analysis of metadata fields reveals key distributional properties from the real dataset:

Pull Request Status Analysis:

- **Overall closure rate**: 81.7% (34,465 closed, 7,714 open)
- **Copilot closure rate**: 81.8% showing consistent patterns
- **Claude Code closure rate**: 80.8% demonstrating similar behavior
- **Status consistency**: Minimal variation between agents (0.9% difference)

Dataset Column Structure: The aidata.csv contains 14 metadata columns including:

- **Core identifiers**: id, number, user, user_id, repo_id
- **Temporal data**: created_at, closed_at, merged_at timestamps
- **Content fields**: title, body with high completeness rates
- **Agent classification**: agent field enabling comparative analysis

Top Contributors Analysis: The dataset reveals concentrated contribution patterns:

- **Primary contributor**: "Copilot" user with 36,596 PRs
- **Secondary contributors**: "randallb" (508 PRs), "claudeai-v1[bot]" (223 PRs)
- **User distribution**: Long-tail pattern typical of open source development

The observed user activity distribution shows extreme concentration patterns.

D. RQ4: Methodological Implications and Recommendations

The dataset analysis provides important guidance for empirical research design:

Sample Size Adequacy: With 42,179 total records distributed across two label categories (37,042 Copilot-labeled, 5,137 Claude-labeled), the dataset provides sufficient statistical power for exploratory structural analysis while maintaining manageable computational requirements, though extreme user clustering limits behavioral generalizations.

TABLE V
DATASET QUALITY ASSESSMENT

Quality Metric	Value
Total records	42,179
Agent balance ratio	7.2:1 (Copilot:Claude)
Data completeness	99.3%
Missing titles	1 (0.002%)
Missing body content	300 (0.71%)
Unique users	1,934
Median PRs per user	1.0
Closure rate	81.7%

User Independence Considerations: The highly skewed user contribution pattern (one user contributing 86.7% of data) requires careful statistical treatment. Researchers should consider:

- User-level clustering effects in statistical models
- Stratified sampling approaches to balance user contributions
- Separate analysis of high-volume vs. occasional contributors

Missing Data Management: The minimal missing data (0.71% body content, 0.002% titles) allows for straightforward complete-case analysis without significant bias introduction.

Label Representation: The 7:1 ratio between Copilot and Claude Code provides numerical representation of both categories, though extreme user-level clustering prevents valid comparative behavioral analysis.

V. DISCUSSION

A. Dataset Characteristics and Research Implications

The empirical analysis of aidata.csv reveals important characteristics that inform empirical research design:

Agent Label Distribution: The 87.8% Copilot vs. 12.2% Claude Code distribution provides numerical representation of both labels, but does not support valid behavioral comparison due to extreme user-level clustering. The 7:1 ratio avoids extreme imbalances, but clustering effects prevent meaningful comparative inference.

User Contribution Patterns: The highly skewed user distribution (one user contributing 86.7% of data) exceeds typical open-source contribution patterns and likely reflects automation or institutional data collection rather than individual developer behavior. This requires careful methodological consideration and limits behavioral interpretations.

Data Quality Adequacy: With 99.3% data completeness and minimal missing fields, the dataset supports robust empirical analysis without extensive preprocessing or imputation requirements.

B. Methodological Considerations for Future Research

The dataset characteristics suggest several important methodological guidelines:

Sample Size Considerations: The 42,179 total records provide large numerical coverage for descriptive and structural analyses while remaining computationally manageable. The binary label distribution simplifies structural analysis while avoiding multi-class complexity.

User-Level Dependencies: The concentrated contribution pattern requires careful treatment of user-level clustering in statistical models. Researchers should consider mixed-effects approaches or user-stratified analysis to account for non-independence.

Temporal and Repository Effects: The dataset's longitudinal nature and multi-repository coverage offer opportunities for studying temporal patterns and repository-level variations in AI tool adoption and effectiveness.

C. Research Opportunities and Guidelines

The real dataset characteristics enable several valuable research directions:

Label Distribution: The 7:1 Copilot to Claude Code ratio provides numerical representation of both categories, though extreme user-level clustering prevents valid comparative behavioral analysis between agents.

User-Stratified Analysis: The concentration of contributions (one user with 86.7% of data) suggests analyzing high-volume vs. occasional users separately to understand different usage patterns and effectiveness measures.

Temporal Metadata: The dataset contains temporal metadata that may support limited structural time-based inspection, though batch automation and clustering constrain behavioral time-series inference.

D. Recommendations for Empirical Research

Based on the empirical analysis, we recommend:

1) **Account for User Clustering**: Use mixed-effects models or user-stratified analysis to handle the concentrated contribution pattern
2) **Leverage Binary Label Structure**: The two-agent structure simplifies structural analysis while providing sufficient contrast
3) **Minimal Preprocessing Required**: The high data completeness (99.3%) allows direct analysis without extensive imputation
4) **Consider Temporal Patterns**: Exploit the longitudinal nature for studying adoption and learning effects

E. Scope and Limitations

Analysis Scope:

- **Complete Dataset Analysis**: We analyzed all 42,179 records from aidata.csv to capture structural patterns that sampling could miss
- **Metadata-Only Assessment**: Analysis focuses on available metadata fields without code content examination
- **Quality Documentation**: We document observed issues without attempting to correct or validate the underlying data

Methodological Constraints:

- **No Behavioral Inference**: This analysis makes no claims about AI agent capabilities, performance, or development practices

- **No Label Validation**: Agent labels are analyzed as provided without independent verification of accuracy
- **Descriptive Focus**: We report patterns and distributions without causal interpretation or comparative claims

Dataset Scope Limitations:

- **GitHub-Specific**: Data reflects GitHub pull request patterns, not broader development practices
- **Collection Period**: Limited temporal span may not represent steady-state usage

F. Future Research Directions

This characterization enables several research opportunities while highlighting methodological requirements:

Label Validation Studies: Manual annotation of subsamples could validate agent attribution accuracy and develop correction methods for the observed inconsistencies.

Code-Level Quality Assessment: Analysis of actual code changes could determine whether metadata quality issues extend to the code content and provide additional dataset characterization.

Methodological Development: This dataset provides a case study for developing robust methods to handle severe imbalance, missing data, and dependency structures in software engineering datasets.

Alternative Dataset Development: The limitations documented here could inform improved data collection strategies for future AI-assisted development datasets.

Preprocessing Framework Development: The specific quality issues identified could guide development of standardized preprocessing pipelines for similar datasets in MSR research.

VI. Threats to Validity

A. Internal Validity

User Clustering Effects: The extreme concentration of contributions (86.7% from one user) creates severe clustering that violates independence assumptions in statistical analysis. This concentration may reflect automated generation rather than diverse human usage, threatening the validity of behavioral inferences.

Temporal Dependencies: PRs may exhibit temporal clustering or batch generation patterns that create dependencies between observations, violating assumptions required for standard statistical approaches.

B. External Validity

Platform Specificity: Data derives exclusively from GitHub, limiting generalizability to other development platforms, version control systems, or organizational contexts where AI tools are deployed differently.

Time Period Limitations: The dataset captures a specific temporal window that may not represent steady-state AI tool adoption or usage patterns across different development phases or organizational maturity levels.

Repository Sampling: The dataset may not represent a random sample of GitHub repositories, potentially biasing results toward specific project types, sizes, or development communities.

C. Construct Validity

AI Agent Attribution: The binary classification (Copilot vs. Claude Code) may not capture the complexity of multi-tool environments where developers use multiple AI assistants or hybrid approaches. Agent labels are taken as provided without independent verification, and the extreme concentration in the ”Copilot” user account may represent system-generated data rather than authentic tool usage.

PR Representation: Pull requests may not accurately represent all AI-assisted development activity, as developers may use AI tools for code that never reaches the PR stage or gets squashed into larger commits.

Quality Metrics: Standard GitHub metadata (closure rates, PR sizes) may not adequately capture AI tool effectiveness or developer satisfaction with AI-generated contributions.

VII. Conclusion

This comprehensive characterization of the official MSR 2026 HuggingFace PR-level release provides essential insights for empirical research design. Analysis of all 42,179 pull requests reveals manageable distributional properties and adequate data quality for robust empirical studies.

Key Findings: The dataset provides adequate field-level completeness for exploratory structural analysis, though extreme user concentration (86.7% from one account) prevents behavioral generalizations about tool effectiveness.

Methodological Implications: The dataset may support exploratory, methodologically constrained analyses after explicitly accounting for extreme user-level clustering. The binary label structure enables limited structural analysis, though the extreme user concentration prevents generalizable behavioral conclusions about tool effectiveness.

Research Contribution: This analysis provides empirically computed dataset characteristics, enabling informed research design and preventing methodological errors based on incorrect assumptions about dataset properties.

Future Directions: The documented characteristics support development of methodologically constrained analyses examining workflow-level and process-level differences under strict controls for user clustering and automation.

Data Availability

The MSR 2026 Challenge dataset is publicly available from HuggingFace (aidata.csv). Our analysis was performed using Python with pandas for data manipulation and matplotlib/seaborn for visualization. The complete analysis script (genuine_analysis.py) and processed results (genuine_analysis_results.json) that generated all statistics in this paper are available at: https://github.com/ghost1100/MSR/

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References

[1] M. Chen et al., "Evaluating Large Language Models Trained on Code," arXiv preprint arXiv:2107.03374, 2021.

[2] J. Austin et al., "Program Synthesis with Large Language Models," arXiv preprint arXiv:2108.07732, 2021.

[3] E. Nijkamp et al., "CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis," *International Conference on Learning Representations*, 2023.

[4] A. Bareiß et al., "Code generation tools in practice: A user study with GitHub Copilot," *Proceedings of the 20th International Conference on Mining Software Repositories*, pp. 285-297, 2023.

[5] C. Bird et al., "Don't touch my code! Examining the effects of ownership on software quality," *Proceedings of the 19th ACM SIGSOFT Symposium and the 13th European Conference on Foundations of Software Engineering*, pp. 4-14, 2011.

[6] S. Kim et al., "Classifying software changes: Clean or buggy?" *IEEE Transactions on Software Engineering*, vol. 34, no. 2, pp. 181-196, 2008.

[7] A. Bacchelli and C. Bird, "Expectations, outcomes, and challenges of modern code review," *Proceedings of the 2013 International Conference on Software Engineering*, pp. 712-721, 2013.

[8] N. Niu, "On the role of testing in software evolution," *IEEE Software*, vol. 39, no. 2, pp. 45-52, 2022.

[9] L. Inozemtseva and R. Holmes, "Coverage is not strongly correlated with test suite effectiveness," *Proceedings of the 36th International Conference on Software Engineering*, pp. 435-445, 2014.

[10] MSR Challenge 2026: AI Development Dataset. Available: https://2026.msrconf.org/track/msr-2026-mining-challenge

[11] GitHub Copilot Documentation and Usage Statistics. GitHub Inc., 2024.

[12] OpenAI, "OpenAI Codex," Available: https://openai.com/codex/, 2024.